

Image Segmentation Based on Visual Prompt

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Abstract—Image segmentation, a cornerstone of computer vision, involves partitioning an image into meaningful segments, a task fundamental to high-level image understanding. This paper presents a comprehensive survey of the evolution of image segmentation techniques. We trace the trajectory from early, pre-deep learning methods that relied on handcrafted features and mathematical models, to the revolutionary impact of the deep learning era, marked by architectures like Fully Convolutional Networks (FCN), U-Net, and Mask R-CNN. We then delve into the modern landscape, which is increasingly dominated by Transformer-based architectures and the paradigm of foundation models. A significant focus is placed on the Segment Anything Model (SAM), a recent breakthrough that introduced the concept of promptable, zero-shot segmentation, aiming to create a single, generalist model for a vast array of segmentation tasks. This survey analyzes the motivations, problem formulations, and core contributions of these pivotal works, providing a structured overview of the field's progress and highlighting the paradigm shift towards generalized, prompt-driven segmentation.

Index Terms—Image Segmentation, Deep Learning, Foundation Models, Segment Anything (SAM), Computer Vision

I. INTRODUCTION

A. General Background

Image segmentation is a fundamental task in computer vision that involves partitioning a digital image into multiple segments or sets of pixels, often corresponding to different objects or parts of objects. The goal is to simplify or change the representation of an image into something that is more meaningful and easier to analyze. The history of this field can be broadly divided into three distinct eras.

The Pre-Deep Learning Era (c. 1970s–2000s): Before the widespread adoption of deep learning, segmentation methods were primarily based on handcrafted features and mathematical models. These techniques required significant domain knowledge and focused on pixel-level properties such as intensity, color, texture, and edges. Prominent methods from this period include:

- **Canny Edge Detection:** Proposed by John Canny in 1986, this method became a standard for detecting a wide range of edges in images through a multi-stage algorithm [1].
- **Active Contour Models (Snakes):** Introduced by Kass et al. in 1988, these deformable models were particularly effective for segmenting objects with irregular shapes, finding extensive use in medical imaging [2].

- **K-Means Clustering:** This unsupervised algorithm, formalized by MacQueen in 1967, partitions pixels into clusters based on feature similarity, serving as a foundational clustering-based segmentation technique [3].

These methods, while foundational, relied on heuristics and low-level features, often struggling with complex scenes and semantic understanding.

The Deep Learning Era (c. 2010s): The advent of deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized image segmentation. These models could automatically learn hierarchical features from data, eliminating the need for manual feature engineering. Key milestones include:

- **Fully Convolutional Networks (FCN):** Long et al. (2015) introduced FCNs, which replaced fully connected layers in CNNs with convolutional layers, enabling end-to-end training for dense, pixel-wise prediction. The use of skip connections allowed the combination of deep, semantic features with shallow, appearance features [4].
- **U-Net:** Proposed by Ronneberger et al. (2015), U-Net features a symmetric encoder-decoder architecture with extensive skip connections. This design proved highly effective for biomedical image segmentation, achieving remarkable performance with very few training images [5].
- **Mask R-CNN:** He et al. (2017) extended the Faster R-CNN object detection framework by adding a parallel branch for predicting segmentation masks. Mask R-CNN became a dominant framework for instance segmentation and a versatile backbone for many downstream tasks [6].

Modern Research Directions: Current research continues to build upon these deep learning foundations, exploring new architectures and learning paradigms for greater accuracy, efficiency, and generalization.

- **Transformer-Based Architectures:** Vision Transformers (ViTs) and self-attention mechanisms have been adapted for segmentation, leading to models like TransUNet (2021) that combine the strengths of Transformers and U-Net [7].
- **Foundation Models:** A recent paradigm shift involves building large-scale, pre-trained models that can be adapted to various tasks with minimal fine-tuning. The Segment Anything Model (SAM) from Meta AI (2023) exemplifies this, offering prompt-based, zero-shot seg-

mentation for images [8]. Its successor, SAM 2 (2024), extends these capabilities to video [9].

- **Self-Supervised Learning:** Techniques like contrastive learning (e.g., SimCLR) are being explored to reduce reliance on large labeled datasets, enabling models to learn powerful representations from unlabeled data.

B. Motivation

The drive to advance image segmentation stems from both scientific inquiry and practical necessity.

Scientifically, segmentation addresses a fundamental question in perception: while image classification tells us *what* is in an image, segmentation answers *where* it is, down to the pixel level. This granular understanding is crucial for emulating the human ability to parse a scene into distinct objects and regions, forming the basis for true visual comprehension.

Practically, segmentation is an enabling technology for a multitude of real-world applications:

- **Autonomous Driving:** Segmenting lanes, pedestrians, vehicles, and traffic signs is critical for safe navigation.
- **Medical Imaging:** Precise segmentation of tumors, organs, and other anatomical structures is vital for diagnosis, treatment planning, and surgical guidance.
- **Real-World Data Complexity:** Segmentation provides a robust way to handle complex scenes with occluded or overlapping objects.
- **Downstream Vision Tasks:** It serves as a foundational step for numerous other tasks, including object detection, tracking, and detailed semantic scene understanding. Without accurate segmentation, the performance of these downstream applications would be severely limited.

The motivation behind the **Segment Anything Model (SAM)** represents a culmination of these goals. As stated by its creators, “In this work, our goal is to build a foundation model for image segmentation” [8]. The aim was not to create another specialized model but a generalist one. They sought “to develop a promptable model and pre-train it on a broad dataset using a task that enables powerful generalization. With this model, we aim to solve a range of downstream segmentation problems on new data distributions using prompt engineering” [8]. This ambition marks a shift towards creating a single, powerful tool that can handle any segmentation request on any image, democratizing the technology and accelerating research and application development.

C. Problem Statement

Image segmentation is the task of grouping pixels that belong to a common semantic object and separating them from other objects and the background. A “semantic object” refers to an entity that is meaningful and recognizable within a given context, such as a car, a person, a tree, or even amorphous regions like “sky” or “road”.

The core principle of segmentation is guided by two criteria: pixels within the same group should be as similar as possible, while pixels belonging to different groups should be as distinct as possible. This “similarity” can be defined by

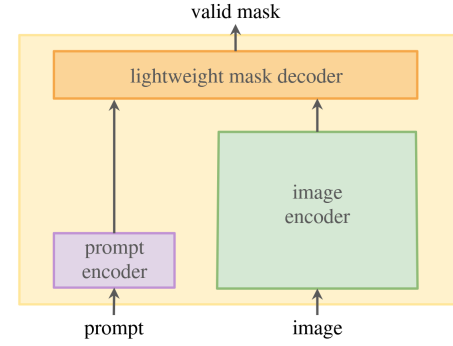


Fig. 1. A general framework for a promptable segmentation model. An image encoder computes a feature embedding for the input image. A prompt encoder embeds the user’s prompt (e.g., points, boxes). A lightweight mask decoder combines these embeddings to predict segmentation masks in real-time.

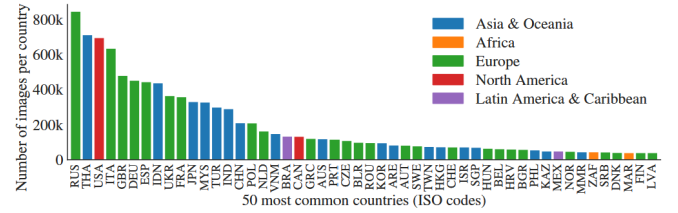




Fig. 3. Diagram for Input/ Output here + Framework etc etc

D. Contribution

The work on the Segment Anything Model (SAM) introduced several pivotal contributions to the field. As summarized by the authors, “Our principal contributions are a new task (promptable segmentation), foundational model for that new task (SAM), and dataset (SA-1B) that make this leap possible” [8].

A key contribution is the **SA-1B dataset**, which remains the largest segmentation dataset to date. Its key features include:

- **Scale:** It contains over 11 million high-resolution images and 1.1 billion high-quality segmentation masks.
- **Privacy-First:** All images are privacy-respecting, with automated blurring of faces and license plates.
- **Diversity:** The dataset is geographically diverse, with a significant number of images from countries around the world, as illustrated in Fig. 2. This helps mitigate geographical bias often present in large-scale datasets.

The contribution of this report is to provide a comprehensive survey and structured analysis of the field of image segmentation. We synthesize the historical evolution, explain the motivations driving modern research, and offer a detailed examination of the methodology behind foundation models like SAM, contextualizing their impact and outlining future directions.

II. RELATED WORKS

A. Traditional Methods

1) **K-Means Clustering:** K-Means is an unsupervised clustering algorithm used to segment regions of interest from the background [10]. It was first introduced by Hartigan in 1975, and its goal is to partition M points in an N -dimensional space into K clusters such that the within-cluster sum of squares is minimized [11].

- 1) **Mechanics:** The algorithm iteratively assigns each pixel to the nearest cluster centroid and updates centroids based on current assignments. K-Means algorithm seeks a local optimum rather than a global one. Data points are divided into groups by assigning each of them to the nearest centroid. The goal is to find a solution such that moving any point from one cluster to another does not reduce the within-cluster sum of squares [11].
- 2) **Methodology:** Let $p(x, y)$ be a pixel at position (x, y) to be cluster and c_k be the centroid of cluster k . Let C_k be the set of pixels assigned to cluster k , where (x, y) indexes the image pixels and $k \in \{1, 2, \dots, K\}$.

For each pixel, compute its distances to all cluster centroids:

$$D_{x,y} = \{ \|p_{x,y} - c_k\| \mid k \in \{1, 2, \dots, K\} \}$$

Assign the pixel to the cluster with the smallest distance:

$$n = \arg \min_k \|p_{x,y} - c_k\|,$$

$$C_n = C_n \cup \{p_{x,y}\}$$

After all pixels are assigned, update each centroid as the mean of the pixels belonging to it:

$$c_k = \frac{1}{|C_k|} \sum_{p_{x,y} \in C_k} p_{x,y}$$

Repeat the assignment and update steps until the centroids no longer change significantly or a maximum number of iterations is reached.

- 3) **Input:** An image represented by M pixels, each with N features (e.g., RGB color components or combined color-spatial features), forming an $M \times N$ matrix. The algorithm also takes K initial cluster centroids in the same N -dimensional feature space.
- 4) **Output:**
 - A label map of size $M \times 1$ indicating the cluster each pixel belongs to.
 - A vector of size $K \times 1$ containing the number of pixels in each cluster.
 - A $K \times N$ matrix of the final cluster centroids.
- 5) **Algorithm:**

```

1  1. Initialize cluster centroids
2  2. For each pixel (x, y):
3     Compute its distance to all centroids
4     Assign it to the cluster with the
5     smallest distance
6  3. For each cluster:
7     Recalculate the centroid as the mean of
8     all pixels assigned to it
9  4. Repeat steps 2 and 3 until convergence

```

2) **Binary Thresholding:** Binary thresholding is used to separate the foreground (region of interest) from the background based on pixel intensity values. It is particularly effective when the objects and background have distinct intensity distributions.

- 1) **Mechanics:** Each pixel in the grayscale image is compared against a threshold value T . Pixels with intensity greater than or equal to T are classified as foreground, while the others are classified as background. This process converts a grayscale image into a binary one.
- 2) **Methodology:** For each pixel (x, y) in the image:
 - a) Compare $I(x, y)$ with the threshold value T .
 - b) Assign the binary value based on the comparison rule.

T can be predefined or determined adaptively using methods such as Otsu’s algorithm, which selects the threshold that minimizes the intra-class variance between the foreground and background.

TABLE I
COMPARISON TABLE - PART 1: METHODOLOGY

Papers	Mechanics	Methodology		
		Stage 1	Stage 2	Stage 3
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TABLE II
COMPARISON TABLE - PART 2: PERFORMANCE AND ANALYSIS

Papers	Performance			Contribution	Limitation
	Data	Metric	Result		
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- 3) **Input:** A grayscale image, where (x, y) denotes the pixel coordinates, $I(x, y) \in [0, 255]$ represents its intensity, a threshold value T to determine the output pixel at (x, y) .
- 4) **Output:** A binary image B , where

$$B(x, y) = \begin{cases} 1, & \text{if } I(x, y) \geq T, \\ 0, & \text{otherwise.} \end{cases}$$

5) **Algorithm:**

```

1  For each pixel (x, y):
2  If I(x, y) >= T:
3    B(x, y) = 1
4  Else:
5    B(x, y) = 0
6
```

B. Modern Methods

1) **Fully Convolutional Networks (FCNs):** Fully Convolutional Networks, introduced by Long et al. in 2015, revolutionized semantic segmentation by adapting classification networks for dense prediction tasks [?]. FCN replaced fully connected layers with convolutional layers, enabling end-to-end learning for pixel-wise segmentation.

- 1) **Mechanics:** FCN transforms classification networks (e.g., VGG, ResNet) into fully convolutional architectures by replacing dense layers with 1×1 convolutions. The network produces coarse feature maps through successive pooling, then upsamples them back to the original image resolution using transposed convolutions (deconvolution). Skip connections combine deep semantic features with shallow appearance features to recover spatial details lost during downsampling.
- 2) **Methodology:** Let $I \in \mathbb{R}^{H \times W \times 3}$ be the input image and f_θ represent the convolutional encoder producing feature maps at multiple scales. The encoder generates hierarchical features:

$$F_i = f_{\theta_i}(F_{i-1}), \quad F_0 = I$$

where F_i represents features at stage i with progressively reduced spatial dimensions.

Upsampling is performed via transposed convolutions g_ϕ :

$$U_i = g_{\phi_i}(U_{i-1})$$

Skip connections fuse features from encoder and decoder:

$$U_i = U_i + W_i \odot F_i$$

where W_i are learned weights and $\odot$ denotes element-wise multiplication.

The final prediction is obtained through a 1×1 convolution:

$$P = \text{softmax}(W_{\text{out}} * U_{\text{final}})$$

- 3) **Input:** An RGB image of arbitrary size $H \times W \times 3$.
- 4) **Output:** A probability map of size $H \times W \times C$, where C is the number of semantic classes. Each pixel is assigned to the class with highest probability.
- 5) **Algorithm:**

```

1  1. Pass image through convolutional encoder
2  2. Store feature maps at multiple scales
3  3. Upsample coarse feature maps via
   transposed convolutions
4  4. Add skip connections from encoder to
   decoder
5  5. Apply 1x1 convolution for class
   prediction
6  6. Output per-pixel class probabilities
7
```

2) **U-Net:** U-Net, proposed by Ronneberger et al. in 2015, features a symmetric encoder-decoder architecture with extensive skip connections [?]. Originally designed for biomedical image segmentation, U-Net achieves high accuracy even with limited training data.

- 1) **Mechanics:** U-Net consists of a contracting path (encoder) that captures context and an expansive path (decoder) that enables precise localization. The encoder applies repeated 3×3 convolutions and 2×2 max

pooling for downsampling. The decoder uses 2×2 transposed convolutions for upsampling, concatenated with correspondingly cropped features from the encoder via skip connections. This architecture allows the network to propagate context information to higher resolution layers.

- 2) **Methodology:** The encoder path consists of repeated application of convolution blocks:

$$F_i^{\text{enc}} = \text{ReLU}(\text{BN}(W_i^{\text{enc}} * F_{i-1}^{\text{enc}}))$$

followed by max pooling for downsampling. The decoder path upsamples and concatenates:

$$F_i^{\text{dec}} = [U_i, \text{crop}(F_{N-i}^{\text{enc}})]$$

where $[,]$ denotes channel-wise concatenation, U_i is the upsampled feature map, and crop ensures spatial alignment.

Each concatenated feature undergoes convolution:

$$F_i^{\text{dec}} = \text{ReLU}(\text{BN}(W_i^{\text{dec}} * F_i^{\text{dec}}))$$

The final segmentation map is generated by:

$$M = \sigma(W_{\text{final}} * F_{\text{final}}^{\text{dec}})$$

where σ is the sigmoid (for binary) or softmax (for multi-class) activation.

- 3) **Input:** An image of size $H \times W \times C$ (typically grayscale or RGB).
- 4) **Output:** A segmentation mask of size $H \times W \times K$, where K is the number of classes.
- 5) **Algorithm:**

```

1  1. Encoder: Apply convolutions and max
2  pooling to downsample
3  2. Bottleneck: Process lowest resolution
4  features
5  3. Decoder: Upsample via transposed
6  convolutions
7  4. Concatenate encoder features via skip
8  connections
9  5. Apply convolutions on concatenated
10 6. Generate final segmentation mask

```

3) *Mask R-CNN*: Mask R-CNN, introduced by He et al. in 2017, extends Faster R-CNN for instance segmentation by adding a parallel branch for predicting segmentation masks [6]. This architecture excels at detecting and segmenting individual object instances.

- 1) **Mechanics:** Mask R-CNN operates in two stages. First, a Region Proposal Network (RPN) generates candidate object bounding boxes. Second, for each region of interest (RoI), the network performs three parallel tasks: class prediction, bounding box regression, and mask prediction. The key innovation is RoIAlign, which preserves spatial alignment during RoI pooling by using bilinear interpolation, crucial for pixel-accurate mask prediction.

- 2) **Methodology:** Let I be the input image and f_{backbone} be the feature extractor (e.g., ResNet-FPN). Feature extraction:

$$F = f_{\text{backbone}}(I)$$

RPN generates proposals:

$$\{B_i, s_i\}_{i=1}^N = \text{RPN}(F)$$

where B_i are bounding boxes and s_i are objectness scores.

For each RoI, RoIAlign extracts aligned features:

$$F_{\text{RoI}} = \text{RoIAlign}(F, B_i)$$

Three parallel heads process F_{RoI} :

$$p_{\text{cls}} = \text{softmax}(W_{\text{cls}} \cdot F_{\text{RoI}}) \quad (\text{classification})$$

$$\Delta B = W_{\text{box}} \cdot F_{\text{RoI}} \quad (\text{box regression})$$

$$M = \sigma(W_{\text{mask}} * F_{\text{RoI}}) \quad (\text{mask prediction})$$

The loss function combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{mask}}$$

- 3) **Input:** An RGB image of arbitrary size, typically resized to have a maximum dimension of 800-1024 pixels.
- 4) **Output:** For each detected instance: class label, confidence score, bounding box coordinates (x, y, w, h) , and a binary mask of size 28×28 (upsampled to the original bounding box size).
- 5) **Algorithm:**

```

1  1. Extract multi-scale features via
2  backbone + FPN
3  2. Generate region proposals using RPN
4  3. For each RoI:
5  a. Apply RoIAlign for spatial alignment
6  b. Predict class probabilities
7  c. Refine bounding box coordinates
8  d. Generate instance segmentation mask
9  4. Apply Non-Maximum Suppression (NMS)
10 5. Output final detections with masks

```

4) *DeepLab*: DeepLab, developed by Chen et al., employs atrous (dilated) convolution and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information without reducing spatial resolution [?]. DeepLabv3+ further incorporates an encoder-decoder structure.

- 1) **Mechanics:** DeepLab uses atrous convolution to expand the receptive field without increasing parameters or reducing resolution. The atrous rate r controls the spacing between kernel weights. ASPP applies parallel atrous convolutions with different rates to capture objects at multiple scales. The encoder-decoder structure (in DeepLabv3+) first extracts features, then gradually recovers spatial details through the decoder with skip connections.
- 2) **Methodology:** Atrous convolution with rate r :

$$y[i] = \sum_{k=1}^K x[i + r \cdot k] \cdot w[k]$$

where x is the input, w is the filter, and y is the output. ASPP applies convolutions at multiple rates:

$$F_{\text{ASPP}} = \text{Concat}(f_{r_1}(F), f_{r_2}(F), \dots, f_{r_n}(F), \text{GAP}(F))$$

where f_{r_i} denotes atrous convolution with rate r_i and GAP is global average pooling.

The decoder upsamples low-level features:

$$F_{\text{dec}} = \text{Upsample}(F_{\text{ASPP}}) + W_{\text{skip}} * F_{\text{low}}$$

Final prediction:

$$P = \text{softmax}(W_{\text{out}} * F_{\text{dec}})$$

- 3) **Input:** An RGB image of size $H \times W \times 3$, typically pre-processed with mean subtraction and normalization.
- 4) **Output:** A semantic segmentation map of size $H \times W$, where each pixel is assigned a class label from C possible classes.
- 5) **Algorithm:**

```

1  1. Extract features using backbone with
2    atrous convolution
3  2. Apply ASPP with multiple dilation rates
4  3. Concatenate multi-scale features
5  4. Upsample in decoder with skip
6    connections
7  5. Apply 1x1 convolution for class scores
8  6. Generate final segmentation map

```

5) *Transformer-Based Segmentation:* Vision Transformers (ViTs) adapt the self-attention mechanism from NLP to computer vision. TransUNet, introduced by Chen et al. in 2021, combines Transformers with U-Net architecture for medical image segmentation [7].

- 1) **Mechanics:** TransUNet uses a CNN to extract initial features, then patches the feature map and processes it through Transformer encoder layers. The self-attention mechanism captures long-range dependencies between image patches, overcoming the limited receptive field of CNNs. The Transformer output is reshaped and fed into a CNN decoder similar to U-Net, with skip connections from the CNN encoder for recovering spatial details.
- 2) **Methodology:** The image is first processed by a CNN encoder:

$$F_{\text{CNN}} = f_{\text{CNN}}(I)$$

Features are divided into patches and flattened:

$$z_0 = [\text{Flatten}(F_1), \text{Flatten}(F_2), \dots, \text{Flatten}(F_N)] + E_{\text{pos}}$$

where E_{pos} are learnable positional embeddings.

Multi-head self-attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where $Q = zW_Q$, $K = zW_K$, $V = zW_V$.

Transformer layer:

$$z'_\ell = \text{MSA}(\text{LN}(z_{\ell-1})) + z_{\ell-1}$$

$$z_\ell = \text{MLP}(\text{LN}(z'_\ell)) + z'_\ell$$

where MSA is multi-head self-attention, LN is layer normalization, and MLP is a feed-forward network.

The decoder upsamples and fuses:

$$M = f_{\text{decoder}}(\text{Reshape}(z_L), F_{\text{CNN}})$$

- 3) **Input:** An image of size $H \times W \times C$, divided into patches of size $P \times P$.
- 4) **Output:** A segmentation mask of size $H \times W \times K$, where K is the number of semantic classes.
- 5) **Algorithm:**

```

1  1. Extract hierarchical features using CNN
2    encoder
3  2. Divide feature map into patches and
4    flatten
5  3. Add positional embeddings
6  4. Process through Transformer encoder
7    layers
8  5. Reshape Transformer output to spatial
9    dimensions
10 6. Upsample through CNN decoder with skip
11   connections
12 7. Generate final segmentation mask

```

6) *Recurrent Neural Networks for Segmentation:* RNNs, particularly Long Short-Term Memory (LSTM) networks, have been applied to image segmentation to model spatial dependencies and context [?]. ReNet and related architectures process images sequentially in horizontal and vertical sweeps.

- 1) **Mechanics:** RNN-based segmentation treats the image as a sequence, processing it row-by-row or with overlapping patches. Four RNN layers sweep the image in different directions (left-to-right, right-to-left, top-to-bottom, bottom-to-top) to capture context from all spatial neighbors. The hidden states from all four directions are concatenated to form a rich contextual representation for each pixel. Multiple ReNet layers can be stacked to increase the receptive field.
- 2) **Methodology:** For a feature map $F \in \mathbb{R}^{H \times W \times C}$, four RNN sweeps compute:

$$h_{i,j}^{\rightarrow} = \text{RNN}(F_{i,j}, h_{i,j-1}^{\rightarrow})$$

$$h_{i,j}^{\leftarrow} = \text{RNN}(F_{i,j}, h_{i,j+1}^{\leftarrow})$$

$$h_{i,j}^{\downarrow} = \text{RNN}(F_{i,j}, h_{i-1,j}^{\downarrow})$$

$$h_{i,j}^{\uparrow} = \text{RNN}(F_{i,j}, h_{i+1,j}^{\uparrow})$$

Hidden states are concatenated:

$$H_{i,j} = [h_{i,j}^{\rightarrow}, h_{i,j}^{\leftarrow}, h_{i,j}^{\downarrow}, h_{i,j}^{\uparrow}]$$

For segmentation, additional layers map to class predictions:

$$P_{i,j} = \text{softmax}(W \cdot H_{i,j})$$

- 3) **Input:** A feature map from a CNN backbone of size $H \times W \times C$, or directly an image if using raw pixels.
- 4) **Output:** A segmentation map of size $H \times W \times K$, where K is the number of classes.
- 5) **Algorithm:**

```

1  1. Extract features using CNN (if
2  applicable)
3  2. Initialize hidden states for four RNN
4  directions
5  3. For each layer:
6  a. Sweep left-to-right, accumulating
7  context
8  b. Sweep right-to-left, accumulating
9  context
10 c. Sweep top-to-bottom, accumulating
11 context
12 d. Sweep bottom-to-top, accumulating
13 context
14 e. Concatenate directional hidden states
15 4. Apply fully connected layer for class
16 prediction
17 5. Output per-pixel class probabilities

```

III. METHODOLOGY

IV. IMPLEMENTATION & EXPERIMENT

V. CONCLUSION

VI. DISCUSSION

A. Limitations

B. Future Work

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